

Rational and Irrational Numbers on the Number Line

Additional Practice 7A.02.05 **ANSWER KEY**

1. Sort the seven numbers into the table. Two of them carry a radical sign and are still rational.

$$0.4 \quad -\sqrt{20} \quad -6 \quad -\sqrt{36} \quad -\frac{2}{3}\pi \quad \sqrt[3]{-27} \quad -\frac{11}{3}$$

Rational	Irrational
0.4	$-\sqrt{20}$
-6	$-\frac{2}{3}\pi$
$-\sqrt{36}$	
$\sqrt[3]{-27}$	
$-\frac{11}{3}$	

$-\sqrt{36} = -6$ and $\sqrt[3]{-27} = -3$ are the two traps. $-11/3$ is printed as a ratio already and needs no work. 20 is not a perfect square, so $-\sqrt{20}$ is irrational; and $-\frac{2}{3}\pi$ is π scaled by a NON-ZERO rational number, which stays irrational — the qualifier matters, because $0 \times \pi = 0$ is rational, and the Reference Sheet states the rule that way.

For questions 2 through 7, compare each pair using $<$, $>$ or $=$.

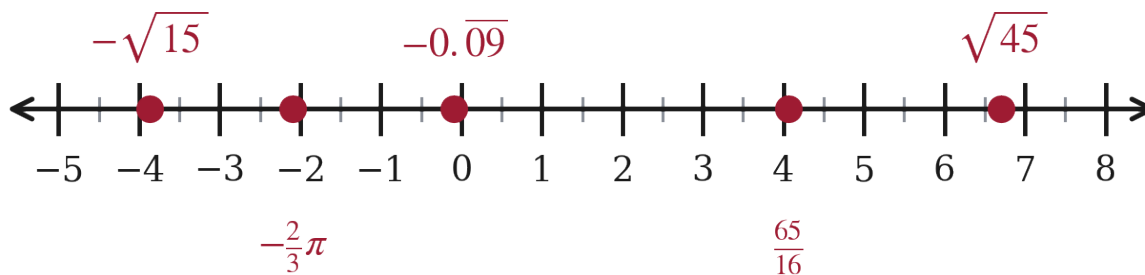
2.	$\sqrt[3]{-56}$	$<$	$-3\frac{3}{4}$
3.	$\frac{\pi}{2} - 3$	$=$	$-3 + \frac{1}{2}\pi$
4.	$\sqrt{20}$	$>$	$4.\overline{23}$
5.	$-\sqrt[3]{125}$	$>$	$-\sqrt{100}$
6.	$\frac{123}{15}$	$<$	8.25
7.	$4\sqrt{35}$	$>$	23.12122...

Question 3 is the equality: $\pi/2 - 3$ and $-3 + \pi/2$ are the same expression written two ways, and a student who reaches for a calculator will still get $=$. Question 5 is the negatives trap — -5 is greater than -10 .

8. Order the numbers from least to greatest, then plot and label each one.

$$-\sqrt{15} \quad -\frac{2}{3}\pi \quad -0.\overline{09} \quad \frac{65}{16} \quad \sqrt{45}$$

$$-\sqrt{15}, -\frac{2}{3}\pi, -0.\overline{09}, \frac{65}{16}, \sqrt{45}$$

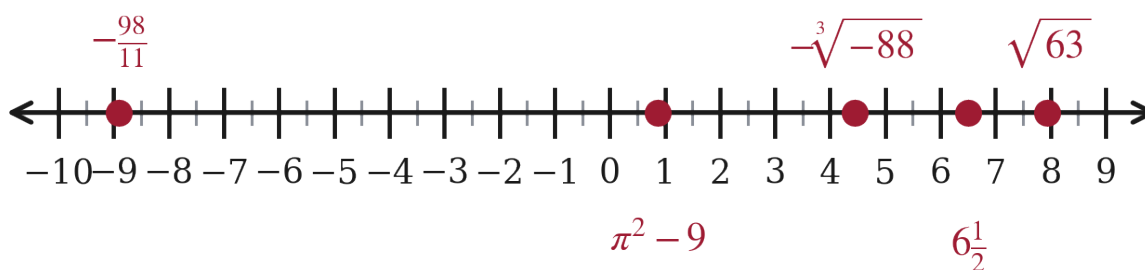


Decimal values: -3.873 ; -2.094 ; $-0.0909\dots$; 4.0625 ; 6.708 . The repeating decimal $-0.090909\dots$ equals $-1/11$. It is the closest of the five to zero, and a student who reads it as -0.9 will put it in the wrong place.

9. Order the numbers from least to greatest, then plot and label each one.

$$-\frac{98}{11} \quad \pi^2 - 9 \quad -\sqrt[3]{-88} \quad 6\frac{1}{2} \quad \sqrt{63}$$

$$-\frac{98}{11}, \pi^2 - 9, -\sqrt[3]{-88}, 6\frac{1}{2}, \sqrt{63}$$



Decimal values: -8.909 ; 0.870 ; 4.448 ; 6.5 ; 7.937 . The third one is a double negative — $\sqrt[3]{-88} \approx -4.448$, so its opposite is positive.

10. Which is greater, $-\sqrt{36}$ or $-\frac{2}{3}\pi$? Explain how you decided, and say what a student who answers the other way is doing wrong.

$-\frac{2}{3}\pi$ is greater. $-\sqrt{36} = -6$ and $-\frac{2}{3}\pi \approx -2.09$, and -2.09 sits to the right of -6 .

A student who answers $-\sqrt{36}$ is comparing 6 against 2.09 and ignoring the sign.

With negatives, the number that looks bigger is the smaller one. Further left on the number line always means less.